THE QUOT FUNCTOR OVER ALGEBRAIC SPACES T

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Allowed standard prerequisites (editorial).

- Situation situation-quot (canonical tag 09TR). cited at statement line 2488. Standing situation for the Quot functor.
- Lemma lemma-extend-q-to-spaces (canonical tag 0D3U). cited at proof line 2508. The ambient identification of Mor with pairs.
- Lemma spaces-descent-lemma-descending-property-proper (canonical tag 0422). cited at proof line 2522. Properness descends along the etale cover.
- Lemma spaces-morphisms-lemma-flat-pullback-support (canonical tag 089C). cited at proof line 2524. Scheme-theoretic support commutes with flat base change.

Source reference. Original definition 09TR: sources/stacks-chow/tags.txt, source lines 12399–12399. Cited at statement line 2488: standing situation.

Source reference. Original lemma 0D3U: sources/stacks-chow/tags.txt, source lines 16429–16429. Cited at proof line 2508: the ambient identification.

Source reference. Original lemma 0422: sources/stacks-chow/tags.txt, source lines 5313–5313. Cited at proof line 2522: properness descends.

Source reference. Original lemma 089C: sources/stacks-chow/tags.txt, source lines 10492–10492. Cited at proof line 2524: support commutes with flat base change.

Lemma 0.2. *In Situation 09TR. Let T be an algebraic space over S. We have*

$$\mathrm{Mor}_{Sh((Sch/S)_{fppf})}(T, \mathrm{Quot}_{\mathcal{F}/X/B}) = \left\{ \begin{array}{c} (h, \mathcal{F}_T \to \mathcal{Q}) \text{ where } h : T \to B \text{ and} \\ \mathcal{Q} \text{ is of finite presentation and} \\ \text{flat over } T \text{ with support proper over } T \end{array} \right\}$$

where $\mathcal{F}_T$ denotes the pullback of $\mathcal{F}$ to the algebraic space $X \times_{B,h} T$.

Proof. Observe that the left and right hand side of the equality are subsets of the left and right hand side of the second equality in Lemma 0D3U. To see that these subsets correspond under the identification given in the proof of that lemma it suffices to show: given $h : T \to B$, a surjective étale morphism $U \to T$, a finite type quasi-coherent $\mathcal{O}_{X_T}$-module $\mathcal{Q}$ the following are equivalent

(1) the scheme theoretic support of $\mathcal{Q}$ is proper over T, and

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(2) the scheme theoretic support of $(X_U \to X_T)^*\mathcal{Q}$ is proper over U.

This follows from Descent on Spaces, Lemma 0422 combined with Morphisms of Spaces, Lemma 089C which shows that taking scheme theoretic support commutes with flat base change. □